

ASSIGNMENT-1

Student's Name _____ Date of Submission: _____

Date: 28-09-2026

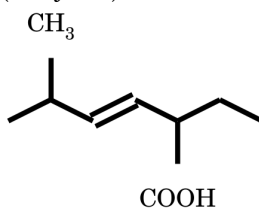
Q1. Let X = number of structural isomers possible for molecular formula C_4H_9Cl ; Y = total number of ether structures possible for $C_4H_{10}O$; Z = total number of structural isomers of C_9H_{18} containing a cyclohexane ring. Calculate X + Y + Z.

Q2. Let X = number of structural isomers possible for $C_4H_8Br_2$; Y = number of isomeric forms of C_7H_9N containing a benzene ring; Z = number of alkynes possible with molecular formula C_5H_8 . Calculate X + Y + Z.

Q3. Let X = total number of structural isomers of $C_2X_2Y_2Z_2$, where X, Y and Z are different monovalent groups; Y = number of structurally different geminal dichlorides possible for $C_5H_{10}Cl_2$; Z = number of structural isomers possible for C_5H_{10} , excluding stereoisomers. Calculate X + Y + Z.

Q4. Total number of isomeric alcohols with molecular formula $C_4H_{10}O$ are _____.

Q5. For the structure shown, x = number of carbon atoms in the principal carbon chain, y = locant of the methyl group and z = locant of C=C. Find $(x + y + z)/2$.

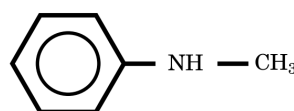
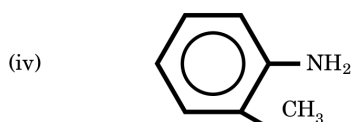
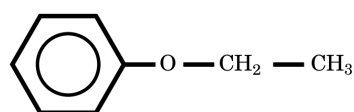
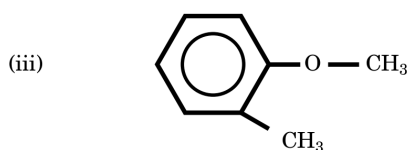
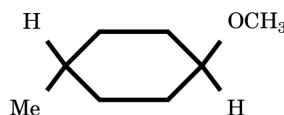
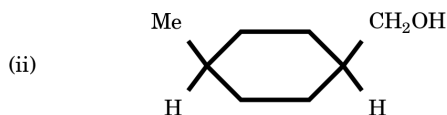
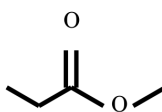
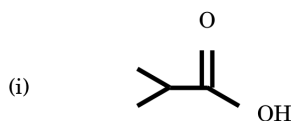


Q6. How many primary amines are possible for molecular formula $C_4H_{11}N$?

Q7. Let X = number of open-chain structural isomers possible for C_4H_8 ; Y = number of structurally isomeric dibromobutanes; Z = number of structurally isomeric dienes possible for C_5H_8 . Calculate X + Y + Z.

Q8. Let X = number of aromatic benzene-ring-containing isomers possible for C_8H_{10} ; Y = number of tetramethylbenzene isomers; Z = number of structural isomers possible for C_6H_{14} . Calculate X + Y + Z.

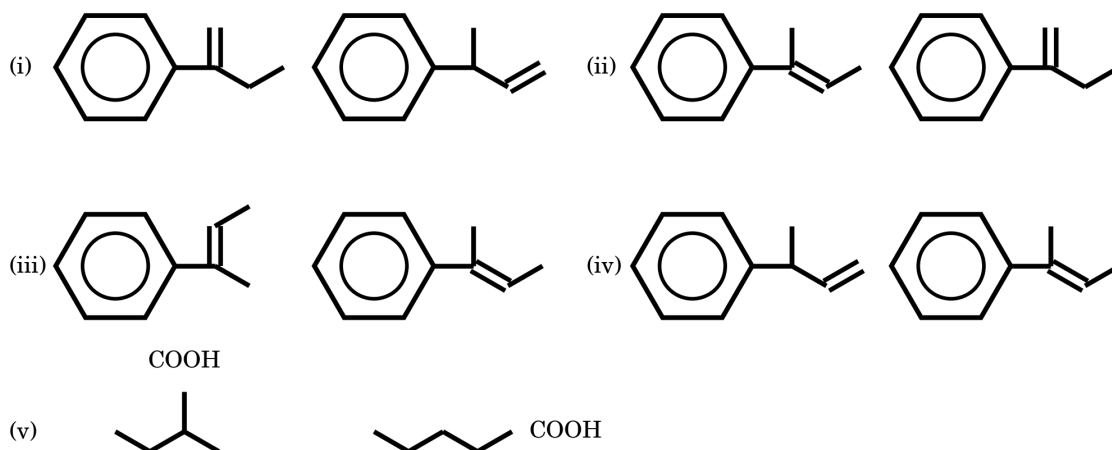
Q9. How many of the following represent pairs of functional-group isomers?



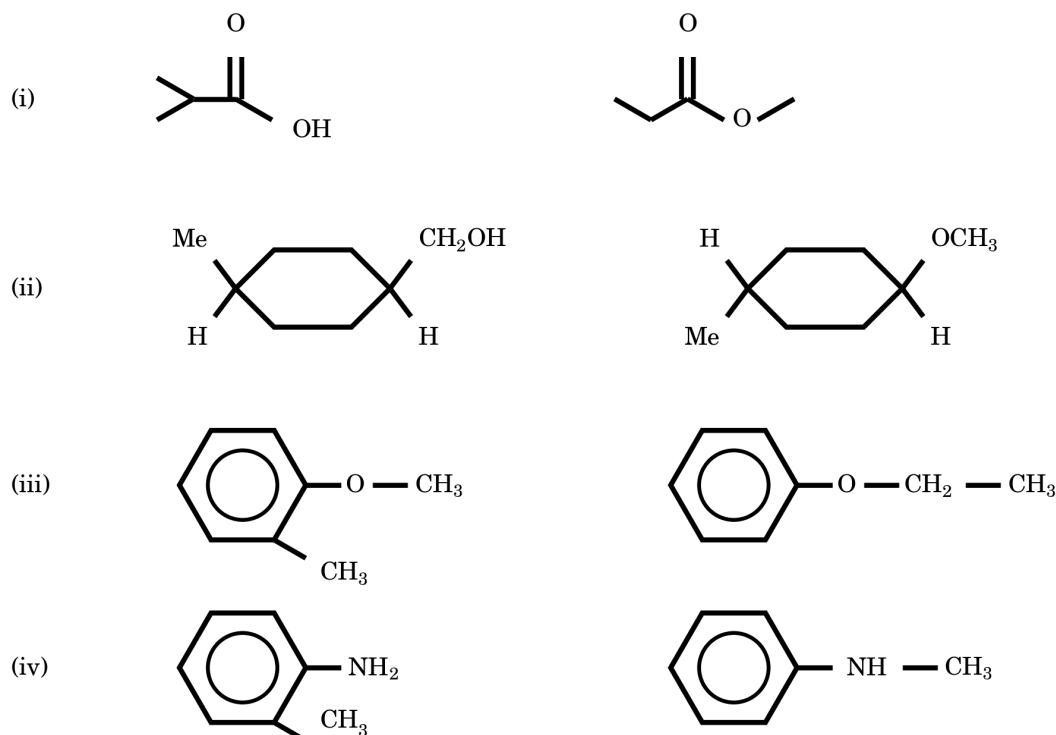
Q10. The number of structural isomers possible for molecular formula $C_2H_2FClBrI$ is ____.

Q11. Let X = number of aldehydes with formula C_4H_8O ; Y = number of chain isomers of C_5H_{12} ; Z = maximum number of alkenes with formula C_4H_8 . Find X + Y + Z.

Q12. How many among the following pairs do not represent positional isomers of each other?



Q13. How many of the following pairs are not functional isomers?



Q14. Let X = total number of structural isomers of tertiary amines having molecular formula $C_6H_{15}N$; Y = number of isomeric forms of C_7H_9N containing a benzene ring; Z = total number of isomers of C_7H_{14} containing only a five-membered ring. Calculate X + Y + Z.

Q15. Let X = number of cyclic ketones possible for C_5H_8O (structural isomers only); Y = number of structural isomers of C_7H_8O containing a benzene ring; Z = number of positional isomers possible for an alkane of molecular mass 72 amu. Calculate X + Y + Z.

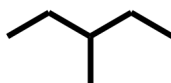
Q16. Let X = minimum molar mass of an alkane that exhibits position isomerism; Y = number of structurally isomeric dibromobutanes containing a four-carbon parent chain. Calculate $X + Y$.

Q17. Let X = number of structural isomers obtained when one H atom of diphenylmethane ($\text{Ph-CH}_2\text{-Ph}$) is replaced by Cl; Y = number of constitutional positional isomers possible for trimethylcyclohexane; Z = number of isomers of xylene. Calculate $X + Y + Z$.

Q18. Let X = number of positional isomers possible for dimethylcyclohexane; Y = number of positional isomers possible for $\text{C}_6\text{H}_3\text{ClBr}_2$; Z = total number of benzene derivatives having molecular formula $\text{C}_7\text{H}_8\text{O}$. Calculate $X + Y + Z$.

Q19. Let X = minimum number of carbon atoms in an ether required to show metamerism; Y = number of metamers possible for $\text{C}_5\text{H}_{12}\text{O}$. Calculate $X + Y$.

Q20. Number of metamers of $\text{CH}_3\text{CH}_2\text{-O-CH}_2\text{CH}_3$ is x (including the given compound). Number of functional isomers of the structure shown is y . Find $x \times y$.



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Q21. Total number of ether structures possible for $\text{C}_5\text{H}_{12}\text{O}$ are ____.

Q22. Spiro compounds have two rings sharing only one carbon atom. How many structurally isomeric spiro compounds, containing a pi bond and with both rings three-membered, are possible for molecular formula C_7H_{10} ?

Q23. How many structurally isomeric bicyclo compounds having at least one three-membered ring are possible for molecular formula C_6H_{10} ?

Q24. The number of different chain isomers for C_7H_{16} is ____.

Q25. How many constitutional (structural) isomers exist for C_6H_6 that contain a cyclopropyl ring?

Q26. Number of possible cyclic alcohols for $\text{C}_5\text{H}_{10}\text{O}$ is ____.

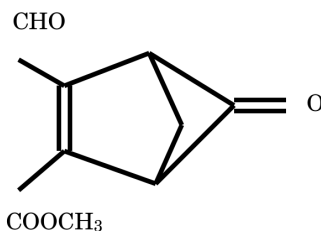
Q27. If the total number of saturated structural isomers corresponding to molecular formula C_5H_8 is n , answer only the n th item from the following: (1) total number of planes of symmetry in benzene; (2) maximum number of atoms in one plane in benzene; (3) total number of geometrical isomers of $\text{CH}_3\text{-CH=CH-CH=CH}_2$; (4) total number of planes of symmetry in open-book H_2O_2 .

Q28. The total number of structural isomers of C_6H_{14} is x . The total number of isomeric alcohols (structural only) with formula $\text{C}_4\text{H}_{10}\text{O}$ is y . The total number of aldehydes (structural isomers) with formula $\text{C}_5\text{H}_{10}\text{O}$ is z . Find $(z + y) - x$.

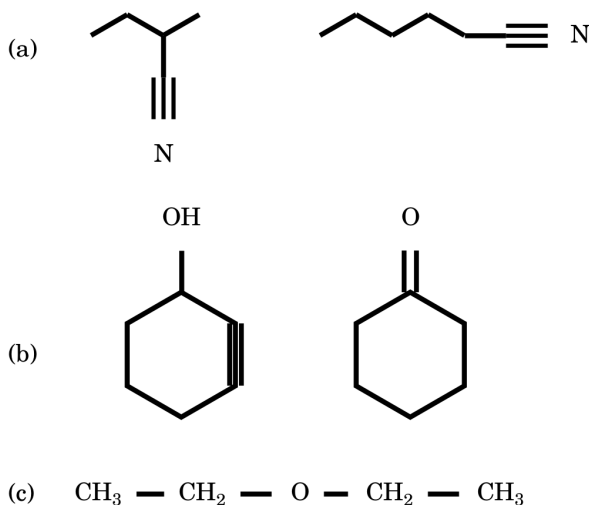
Q29. P is the number of structural isomeric amines with no chiral carbon that give the isocyanide test, and Y is the total number of structural isomeric amines that do not give the isocyanide test, for molecular formula $\text{C}_8\text{H}_{11}\text{N}$. Every isomer must contain a phenyl group. Find $|P - Y|$.

Q30. Dicyclo compounds possible for $\text{C}_5\text{H}_8 = x$. Total cyclic structural isomers possible for $\text{C}_5\text{H}_{10} = y$. Find $(y - x)^2$.

Q31. Find the degree of unsaturation of the compound shown.



Q32. Analyse the three pairs shown. Use: 1 = tautomers, 2 = metamers, 3 = positional isomers, 4 = functional-group isomers, 5 = chain isomers, 6 = not isomers. Write the three digits in the order (a), (b), (c).

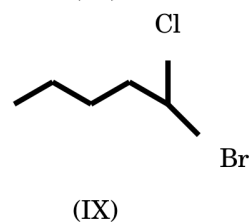
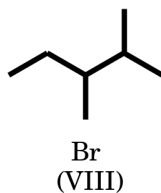
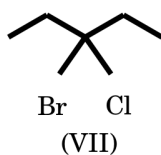
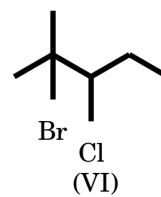
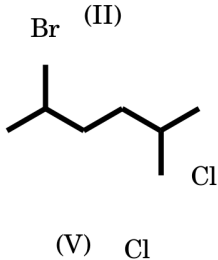
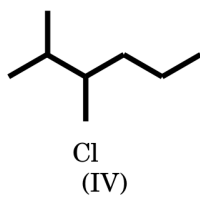
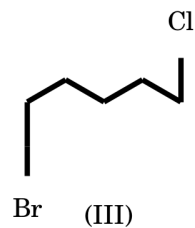
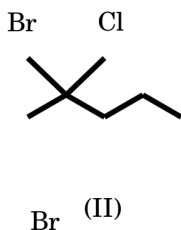
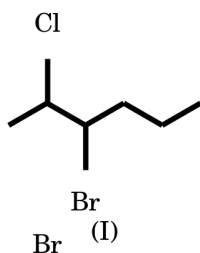


Q33. For molecular formula $\text{C}_3\text{H}_6\text{O}$, let x be the total number of structural isomers, excluding enol forms. Find $4x/5$.

Q34. How many of the following molecular formulae/compounds can show ring-chain isomerism: C_3H_8 , C_4H_8 , C_2H_6 , C_2H_4 , $\text{C}_3\text{H}_6\text{O}$, C_2H_6 , $\text{C}_5\text{H}_5\text{N}$ and $\text{C}_7\text{H}_8\text{O}$?

Q35. The number of structural isomers possible for $\text{C}_6\text{H}_6\text{O}_3$ containing a benzene ring (do not consider peroxide-bond-containing isomers) is X. The number of structural isomers possible for $\text{C}_7\text{H}_9\text{N}$ containing a benzene ring is Y. Find $X + Y$.

Q36. How many of the following compounds are positional isomers of 2-bromo-3-chlorohexane?



ANSWER KEY

Q. No.	Answer	Q. No.	Answer
1	19	19	10
2	17	20	0
3	21	21	6
4	4	22	10
5	7	23	10
6	4	24	9
7	15	25	5
8	12	26	9
9	3	27	2
10	7	28	3
11	9	29	4
12	2	30	4
13	1	31	6
14	16	32	516
15	11	33	$28/5 = 5.6$
16	92	34	4
17	13	35	8
18	15	36	4 (I, III, V, IX)